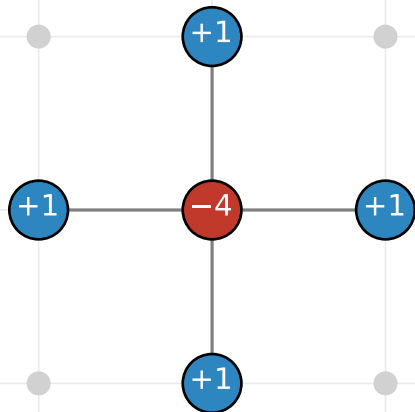


(a) Exchange: 5-point Laplacian

$$\mathbf{H}_{\text{ex}} = \frac{2A_{\text{ex}}}{M_s a^2} (\sum_{\text{nn}} \mathbf{m}_{\text{nn}} - 4 \mathbf{m}_0)$$



$\longleftrightarrow$
 a

(b) Interfacial DMI: central difference

$$\mathbf{H}_{\text{DMI}} = \frac{D}{M_s a} [(m_z^{+\hat{x}} - m_z^{-\hat{x}}) \hat{x} + (m_z^{+\hat{y}} - m_z^{-\hat{y}}) \hat{y} - (\partial_x m_x + \partial_y m_y) \hat{z}]$$

